FinTrust Digital Bank — Week 2 Business Findings

AnalystLab Africa Experience Lab · Data Analytics Track · Week 2 Part E

Structure: Finding → Evidence → Business Meaning. All figures are computed from the cleaned FinTrust datasets produced in Part A.

Item	Detail
Project	FinTrust Financial Intelligence & Digital Banking Support Solution
Week / Version	Week 2 — Analyse & Prepare · v1.0
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Track	Data Analytics

1. Value and review workload sit on different transaction types

Evidence

Transfers carry the highest (synthetic) risk-review rate at 28.49% on 3,549 transactions (avg NGN 67,038), while Deposits carry the highest average value at NGN 98,633 but a below-average review rate of 16.19%. Transfer and Cash Withdrawal together hold 1,373 of 2,352 flagged transactions (58%) on only 41% of volume.

Business Meaning

A review workflow targeted at Transfers and Cash Withdrawals would cover most of the flagged population while allowing fast, low-risk types (Bill Payment 12.81%, Card Purchase 13.02%) to proceed without friction. This converts a blanket-review queue into a prioritised one.

2. International transactions are reviewed at nearly double the rate

Evidence

480 international transactions (4.00% of volume, NGN 19,995,782) show a 36.88% review rate versus 18.88% for the 11,520 domestic transactions — a 1.95x lift.

Business Meaning

A specific international-transaction rule concentrates review effort on 4% of volume instead of applying blanket scrutiny. Because the segment is small, the operational cost of deeper review is low relative to the coverage gained.

3. Night-time transactions are the strongest behavioural signal

Evidence

Transactions between 00:00 and 05:00 are reviewed at roughly 28% versus roughly 17% during the day — a 1.70x difference. High-value transactions (top decile) show a 2.02x lift (36.00% vs 17.78%).

Business Meaning

Two low-cost, high-precision rules emerge: route night-time transactions to a deeper check, and treat top-decile amounts as inherently higher-attention. Because both conditions are a minority of volume, neither throttles normal throughput.

4. Channel is a weak differentiator; absolute failure is not

Evidence

Review rates across channels span only 4 points (Web 21.35%, ATM 21.18%, Mobile App 19.40%, POS 18.35%, USSD 17.32%). Absolute failures,

however, concentrate on volume: Mobile App 296 and POS 125 of the 630 failed transactions (67%).

Business Meaning

Channel-based risk rules should not be adopted — the spread is too narrow to justify them, and the earlier statistical test found channel only marginally significant. Operational investment should instead follow absolute failure counts, where Mobile App and POS offer the largest reduction in customer pain per fix.

5. Value-based customer ranking beats segment-based ranking

Evidence

The single highest-value customer (NGN 1,611,209 across 17 transactions) sits in the Everyday segment, not Premium. Across the top 20 customers by value, Everyday and Student customers appear alongside Premium and SME.

Business Meaning

Segment labels alone would misdirect attention. Ranking customers by realised transaction value identifies a different, more commercially relevant group — supporting value-based rather than label-based relationship management. Names are synthetic labels; Customer_ID is the identifier.

6. Amount skew distorts averages and must be handled explicitly

Evidence

Amount_NGN has skewness 3.2, a mean of NGN 46,706 and a median of only NGN 10,306. 1,474 transactions (12.28%) exceed the upper IQR bound of NGN 117,922.86; none are negative or zero.

Business Meaning

Reporting average transaction value without the median overstates the typical customer transaction by roughly 4.5x. All amount-based dashboards and models should show the median alongside the mean, and modelling must use log1p(Amount). Outliers are genuine values and were retained.

7. A data-quality trap would have corrupted geographic analysis

Evidence

Transaction.Location and Customer.City share the same 8 labels but agree only 12.7% of the time. 192 cells are missing (Device_Type 96, Location 96). Customer_Name repeats (180 unique names for 1,500 customers).

Business Meaning

Any analyst joining Location to City would mis-attribute ~87% of transactions geographically. Fields were kept separate, missing values were imputed as an explicit 'Unknown' category rather than dropped (preserving all 12,000 records), and Customer_ID — not Customer_Name — is used as the join key.